

Khwaja Yunus Ali University

School of Science and Engineering
Department of Computer Science and Engineering

Mid-term Examination: Spring Semester 2020

Program: B.Sc. in CSE (Batch – 9th) 1st Year 2nd Semester

Course Title: Physics - II

Course Code: PHY 1201

Time: 15 Minutes

Total Marks: 15 X 1 = 10

Student ID:	Batch:	Session:
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Part 1: MCQ

These are single best answer type of questions. Each of the following question or statement has five suggested answers or completions. Write down the most appropriate one on your answer script.

- What is the thermometric property of the optical pyrometer?
a) Heat energy b) Light energy c) Brightness d) Resistance e) Volume
- When heat is supplied to a gas of volume 100m^3 at normal pressure, the volume changes to 100.2m^3 . How much work done by the gas?
a) 40460J b) 30360J c) 20200J d) 10160J e) 0.0060J
- "Production of heat by friction" is the example of process.
a) reversible b) irreversible c) adiabatic d) isothermal e) isochoric
- Average energy associated with each independent degree of freedom per mole of gas is
a) $\frac{1}{2}RT$ b) $\frac{1}{2}KT$ c) $\frac{3}{2}RT$ d) $\frac{3}{2}KT$ e) $\frac{5}{2}RT$
- What is the unit of the mean free path?
a) m^3 b) m^2 c) m d) m^{-1} e) m^{-2}
- Efficiency of the Carnot's engine is
a) 40% b) 50% c) 60% d) 80% e) 100%
- "Internal energy depends only on the temperature but not on pressure or volume" – stated by
a) Joule b) Brown c) Sadi Carnot d) Mayer e) Van der Waal's
- How many steps included in the Carnot's cycle?
a) 2 b) 3 c) 4 d) 5 e) 6
- What is the radius of the 1st orbit of hydrogen atom?
a) $0.53\text{\AA}$ b) $1.2\text{\AA}$ c) $2.4\text{\AA}$ d) $3.3\text{\AA}$ e) $5.3\text{\AA}$
- The angular momentum of an electron in the second orbit is equal to
a) $\frac{n}{\pi}$ b) $\frac{\pi}{n}$ c) $\frac{h}{2}$ d) $\frac{\pi}{2}$ e) $\frac{h}{\pi}$
- The degree of freedom for the linear motional body is
a) two b) three c) four d) five e) six
- Which one is the relevant term with thermoelectric thermometer?
a) Brightness b) Heat Energy c) Pressure d) Volume e) Thermo-couple
- What is the unit of the molar specific heat?
a) JK^{-1} b) JCal^{-1} c) KJ^{-1} d) CalJ^{-1} e) J
- The difference between two molar specific heat is equal to
a) dQ b) PV c) R d) 1 e) RT
- How many steps in the Carnot's cycle?
a) 2 b) 3 c) 4 d) 5 e) 6

Part 2: SAQ

Time: 45 Minutes

Total Marks: 15

[Answer any 03 (three) questions from the followings]

[The figures in the right margin indicate full marks for the respective question]

- What is elastic collision? [1]
 - Draw the block diagram of heat engine. What may be the conditions to get an engine of 100% efficiency? [2+2]
- "The entropy is constant in reversible process" – Justify the statement? [5]
- What is the energy of an electron in the 2nd orbit of hydrogen atom? [2]
 - Calculate the wave number for Balmer series. [3]
- What is the principle of temperature measurement? [2]
 - What are the factors on which the mean free path of a gas depends? [3]